



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
MONDAY TEST

CLASS: IX
SUBJECT: MATHEMATICS

FULL MARKS: 40
DATE: 04.12.23

Instructions:

- This paper consists of two printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION: A

1. Choose the correct option

[1 × 5 = 5]

i) The area of an equilateral triangle is $16\sqrt{3} \text{ m}^2$, the length of its side is:

- a) 12 m b) 16 m c) 14 m d) 8 m

ii) The total surface area of a cube with edge 5 cm will be:

- a) 30 cm^2 b) 35 cm^2 c) 150 cm^2 d) 180 cm^2

iii) The area of the rhombus with diagonals $(a + b) \text{ cm}$ and $(a - b) \text{ cm}$ will be:

- a) $(a + b)^2$ b) $(a - b)^2$ c) $\frac{1}{2}(a^2 - b^2)$ d) $\frac{1}{2}(a^2 + b^2)$

iv) In a right-angled triangle, if sum of the squares of the sides making right angle is 289 cm^2 , then the length of the hypotenuse is:

- a) 17 cm b) 16 cm c) $13\sqrt{3} \text{ cm}$ d) $12\sqrt{2} \text{ cm}$

v) Assertion (A): In a parallelogram each diagonal always bisects the parallelogram into two equilateral triangles.

Reason (R): ABCD is rhombus, in which $\angle DAB = 110^\circ$, then the measure of $\angle BDC$ is 35° .

- a) (A) is true but (R) is false b) (A) is false but (R) is true
c) Both (A) and (R) are true d) Both (A) and (R) are false

SECTION: B

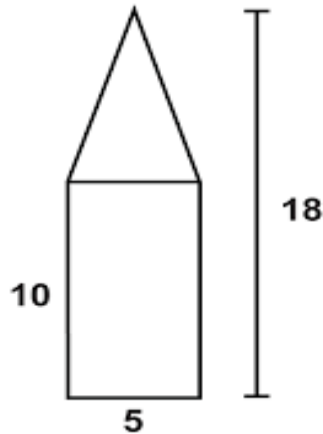
2. In an isosceles triangle ABC with $AB = AC$, BD is perpendicular from B to the side AC. Prove that $BD^2 - CD^2 = 2CD \times AD$.

[4]

3. PQRS is a parallelogram. If the bisector of $\angle P$ and $\angle R$ meet the diagonal QS at M and N respectively, prove that the quadrilateral PMRN is a parallelogram. [4]

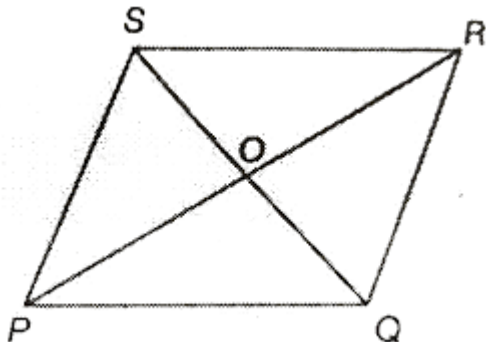
4. The adjacent sides of a parallelogram are 24 cm and 10 cm. If the distance between the larger side is 8 cm find the distance between the shorter sides. Give the answer in nearest cm. [4]

5. The cross section of a water tank consists of a rectangle and an isosceles triangle as shown in the figure. If the tank is 20 m deep, find the volume of water it can hold when full. [all dimensions in m] [4]



6. Three solid cubes, each of volume 216 cm^3 are joined end to end, find the total surface area of the cuboid so formed. [4]

7. In the given figure PQRS is a rhombus, in which $PO = 3x + 7$, $OR = 5x - 3$ and $OQ = 4$ unit. Find the value of x and also perimeter of the rhombus. [use $\sqrt{5} = 2.24$] [5]



8. The perimeter of a triangle is 48 cm and the ratio of its sides are 13: 12: 5. Find the area of the triangle. [5]

9. The external dimensions of a closed wooden box are 18 cm, 15 cm and 13 cm. If the thickness of the used wood is 2 cm and it costs ₹1.50 per cubic cm, find the total cost of the wood required to make the box. [5]